



Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI

Dataset at a Glance

3,900

Transactions

18

Features

37

Missing Values

Review Rating column

\$59.76

Avg. Purchase

Data Preparation in Python

Clean

Imputed missing Review Ratings by category median

Standardize

Renamed columns to snake_case

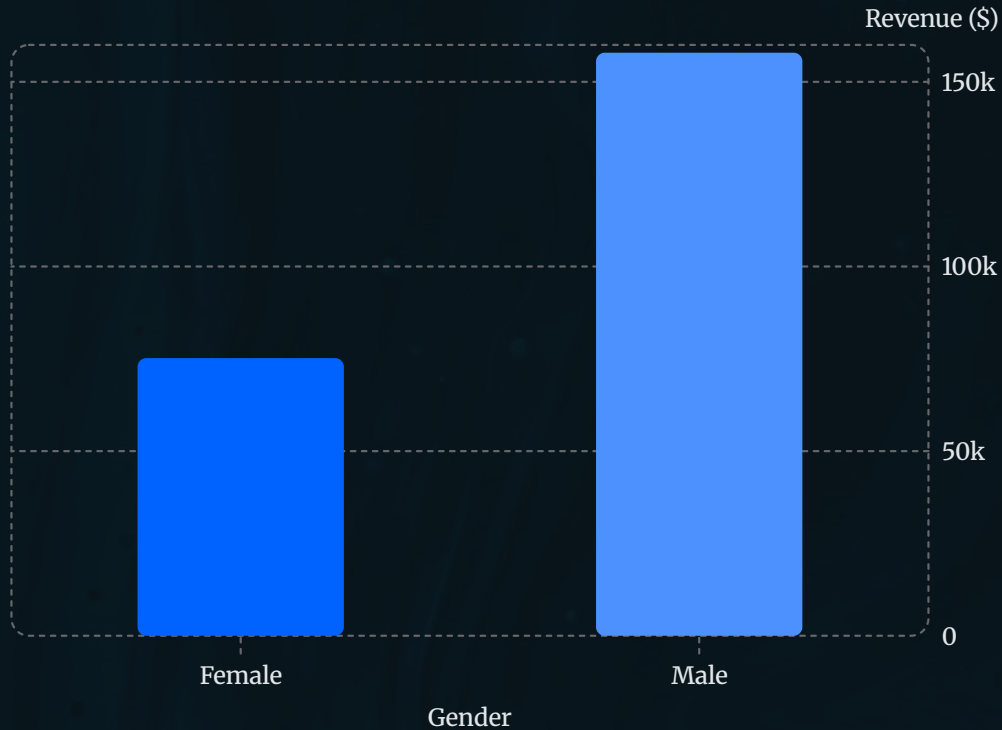
Engineer

Created age_group & purchase_frequency_days

Integrate

Loaded cleaned DataFrame into PostgreSQL

Revenue by Gender



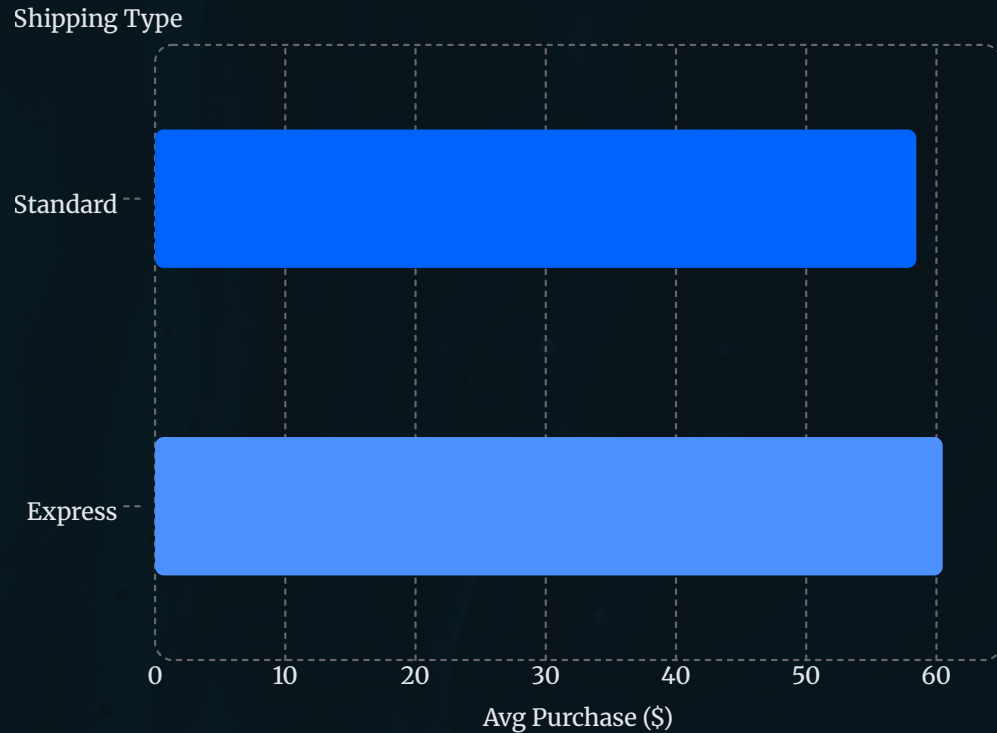
Key Insight

Male customers generated **2× more revenue** than female customers.

Male: **\$157,890**

Female: **\$75,191**

Shipping & Subscription Insights



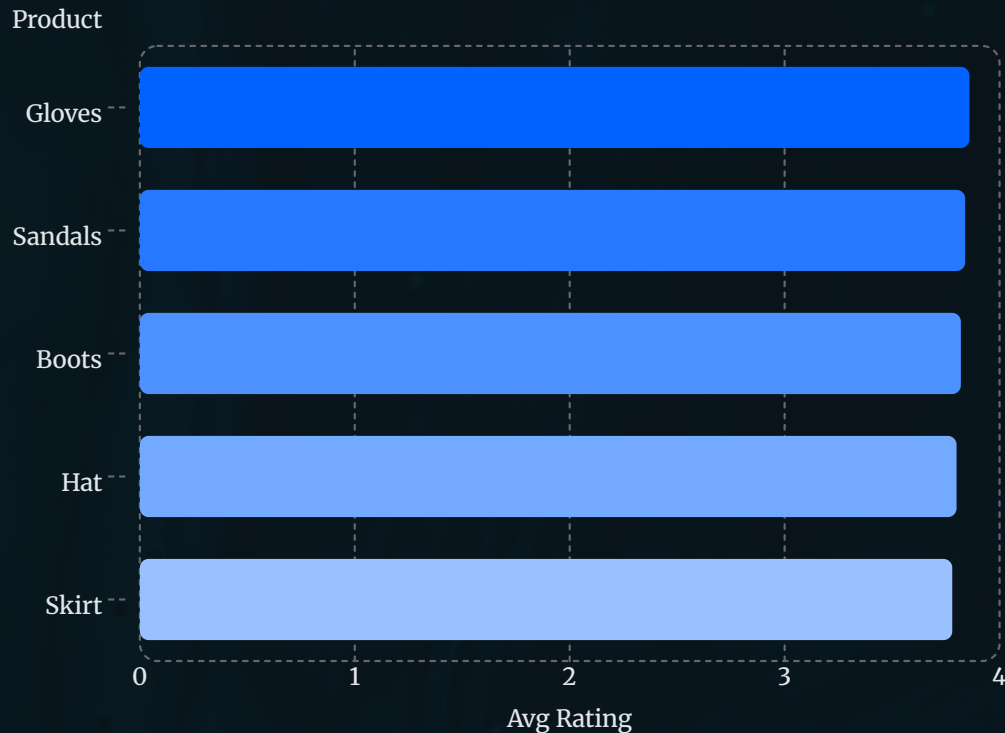
Subscribers vs. Non-Subscribers

Non-subscribers: 2,847 customers · \$170,436 revenue

Subscribers: 1,053 customers · \$62,645 revenue

Express shipping buyers spend \$2 more on average.

Top Products by Rating & Discount Rate



Most Discounted Products

Hat — 50%
discounted
purchases

Sneakers — 49.66%

Coat — 49.07%

Sweater — 48.17%

Pants — 47.37%

Customer Segmentation

3,116

Loyal

80% of base

701

Returning

18% of base

83

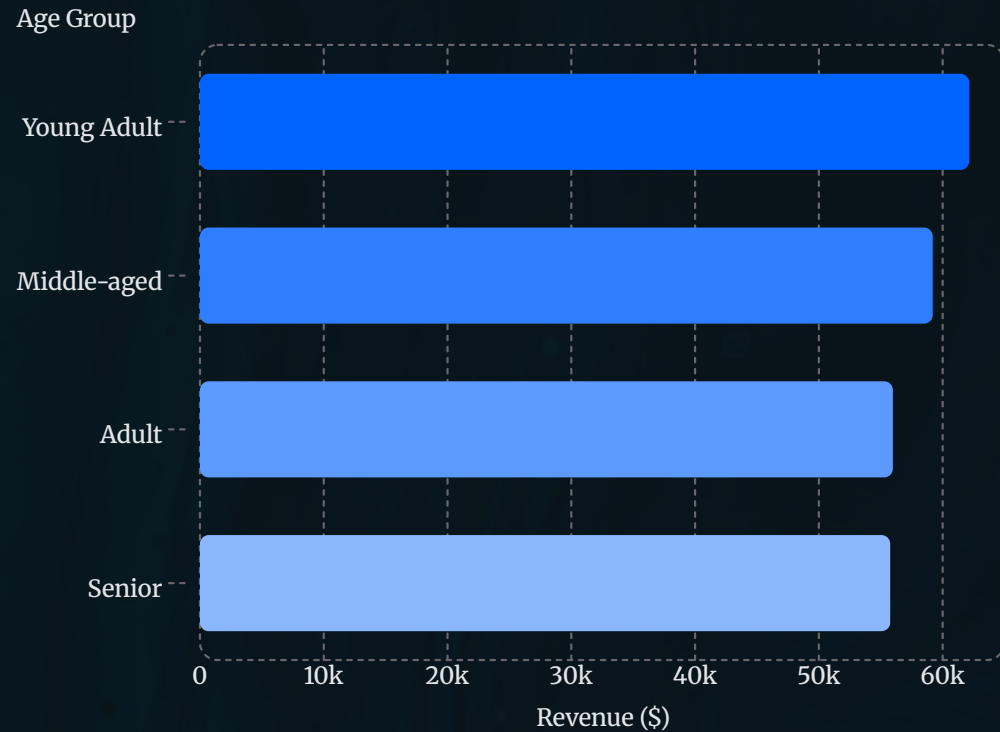
New

2% of base

Of customers with >5 purchases: 2,518 non-subscribers vs. 958 subscribers — repeat buyers are more likely to subscribe.



Revenue by Age Group



Key Insight

Young Adults lead revenue at \$62,143. Revenue is relatively balanced across all age groups.

Power BI Dashboard



Dashboard Highlights

- 3.9K customers · Avg. spend \$59.76 · Avg. rating 3.75
- Clothing leads revenue (\$100K) and sales (1,800 units)
- 27% subscribers vs. 73% non-subscribers
- Filters: Subscription, Gender, Category, Shipping

Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers to grow "Loyal" segment



Review Discount Policy

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express-shipping users

